Ingredient-Based Recipe Recommendation Using Machine Learning

Track: B
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Problem Statement

Oftentimes, household cooks or those who cook with ingredients at home find themselves with leftover ingredients once they finish cooking their main meal. This leftover food is sometimes thrown out which results in a waste of food overall. Even if someone wanted to cook with their leftover ingredients, websites that contain recipes require users to search for a specific dish rather than what ingredients they have to work with. Along with this, we have an information gap with recipes and ingredients. Many recipes are reliant on categories rather than ingredient-based reasoning. With this, people have trouble determining what to cook exactly. We want to create an application that allows users to input their leftover ingredients and be shown the recipes that they can create with what they have, which can help users cook more efficiently and reduce food waste.

Related Work

Work that we were able to find that closely resembled our application was the Recipe Recommendation System (RRS) and RecipeGPT. The RRS was made by Vipanchi Prasanna, it is an ML retrieval system that will recommend recipes by matching the user's ingredient list to recipes in a database. The model uses KNN to identify the most similar recipes based on Euclidean distance. The KNN model is trained on the combined feature set which enables it to recommend recipes by comparing similarities in numerical and textual data.

RecipeGPT was developed by a team of researchers including Helena H Lee, Ke Shu and more. RecipeGPT was designed to automatically generate cooking recipes using AI. It was able to generate ingredient lists and cooking instructions. The text generation model is a pre-trained GPT-2 fine-tuned on large cooking recipe datasets. Unfortunately, the url given doesn't exist anymore, but the paper provides information on the pipeline and how it worked specifically. It was also able to save past recipes generated and compare human-written recipes to use as a reference. According to the paper, it was trained using Tensorflow and deployed to an inference engine to handle live requests. The API service is built on Nginx and UWSGI, MongoDB is then used as data storage to save the generated output for future reference.

Dataset

The dataset we are planning on working with is: link. It contains 180k recipes from Food.com with the list of both raw & transformed data. The raw data includes the full recipe name, ingredients to make it, and more. Format of the transformed data are IDs within a CSV that indicate either the recipe name, which ingredients are included as a list of ids (can be shared among many recipes), steps as a list of ids, and calorie level as a range.

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Approach

For our task, we want a model to be lighter and respond relatively quickly. With that in mind, the team will as of now select the Gemma 2B model. If we learn eventually that a model with more parameters is needed, we can switch to Gemma 7B. Additionally, for training these models on a single GPU we will utilize QLoRA 4-bit precision as that is common.

Baselines

The baseline we plan to have is zero-shot prompting with the base Gemma 2B model, where no fine-tuning has been applied. We will give it a list of ingredients and prompt it to output a recipe and measure the quality of the output. This helps establish what the model can do without any training.

Evaluation

The primary metric to use will be BLEU (Bilingual Evaluation Understudy Score), where it measures n-gram overlap between the model's generated recipe text and the ground-truth recipe. This fits as the output will be a structured text sequence of ingredient list and steps that comes from reference in a datasheet. Thus, it will allow us to compare correct ingredients and instructions with the generation from Gemma.

Ablation Plan

We plan to alter the QLoRA rank, between 4 vs 16 vs 64. Here, we expect to learn the tradeoff between adapter capacity and performance, as rank 4 trains the fastest, but can underfit the complex recipe generation, while rank 64 uses more memory

Compute Plan

Criterion	Detail
Hardware	Google ColabT4
Batch size	4–8 (with gradient accumulation steps of 4 to simulate effective batch size of 16–32)
Estimated training time per run	~1.5–2.5 hours for Gemma 2B on a subset of ~50k recipe samples at QLoRA 4-bit
Number of runs	~5 total: 1 zero-shot baseline, 1 TF-IDF baseline, 3 QLoRA runs (rank 4, 16, 64)

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Team Responsibilities

For week 1, the group will focus on getting the data ready for our model to train on. Specifically, we can use Python with Pandas to manipulate the dataset to obtain our training, testing, and validation splits. Tyler Takimoto and Lucio will work on the week 1 deliverables. To ensure a good base, we will set up a Google Colab environment, as well as a GitHub Repository that we can all access. Also, we will run a simple untuned model to verify that we can get responses. Assigned to week 2 deliverables will be Tyler Tran and Alfred. By week 3, we will be running a QLoRA training script to start our model training. Lucio and Tyler Tran will take the charge for week 3's tasks. Once we have finished training a model, week 4 will start the ablation plans to alter our QLoRA to see how the model responds. Additionally, we will get a basic input functionality set up in a python terminal which will be completed by Alfred and Tyler Takimoto. The team will start to evaluate each of the models, report our findings, and select the best model that we trained. Evaluation will be carried out by the whole team so all members can provide their input on the models. Once we get to the 6th and final week, the whole team will begin creating the final report. At this point, we know all of the information needed to fill out all necessary fields within the report and will have tests to prove our working model.